

LECTURE NOTES MATH255

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1. LECTURE 20

In this class we define splitting fields, normal extensions and finally the Galois group.

1.1. Splitting fields and normal extensions. Let K be a number field $f(x) \in K[x]$ a polynomial with coefficients in K .

Definition 1.1. Let E/K be some extension of K , we say that $f(x) \in K[x]$ splits in E if all zeros of $f(x)$ are elements of E . Equivalently, we say that $f(x)$ splits in E if there is $c \in K$ and $\alpha_1, \dots, \alpha_n \in E$ such that:

$$(1.1) \quad f(x) = c(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n).$$

If $f(x) \in K[x]$ is any polynomial with coefficients in K then we can define a field extension of K by simply adding the zeros of K , this is then a field in which $f(x)$ will split. We call this the splitting field for $f(x)$.

Definition 1.2. Let K be a number field $f(x) \in K[x]$ a non-constant polynomial with coefficients in K , and let $\alpha_1, \dots, \alpha_n$ be the zeros of $f(x)$. We call $K(\alpha_1, \dots, \alpha_n)$ the splitting field of $f(x)$.

Definition 1.3. Let K be a number field and let E/K be a finite extension of K . We say that E is a finite normal extension of K if every irreducible $f(x) \in K[x]$ over K that has at least one zero $\alpha \in E$ splits in E . Equivalently, if $\alpha \in E$ then every conjugate of α is also in E .

Remark 1. In the context of number field, finite normal extensions of K are sometimes called Galois extensions of K . For general fields, we need to add more assumptions on the extension for an extension to be Galois.

With the definition of finite normal extensions we can define the Galois group.

Definition 1.4. Let K be a number field and E/K a finite normal extension of K . We define the Galois group of the field extension E/K as: $\text{Gal}(E/K) = \text{Aut}(E/K)$.

We have the following very important lemma, which will allow us to study zeros of polynomials through the Galois groups of their splitting fields.

Theorem 1.1. Let K be a number field, and let $f(x) \in K[x]$ be a non-constant polynomial. Let $E_f = K(\alpha_1, \dots, \alpha_n)$ be the splitting field of $f(x)$, where $\alpha_1, \dots, \alpha_n$ are the zeros of $f(x)$. Then E_f is a finite normal extension of K .

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Proof. We already know that E_f is a finite extension of K , the only thing we need to prove is that if $g(x) \in K[x]$ is irreducible over K and has a zero in E_f then $g(x)$ splits in E_f . To this end, let $g(x) \in K[x]$ be irreducible over K , and let $\alpha \in E_f$ be a zero for $g(x)$. For any conjugate β of α we find an isomorphism $\psi_{\alpha,\beta} : K(\alpha) \rightarrow K(\beta)$. Since $K(\alpha) \leq E_f$ the extension theorem from last class allows us to extend $\psi_{\alpha,\beta}$ from $K(\alpha)$ to some $\phi : E_f \rightarrow E'$ where $\phi|_{K(\alpha)} = \psi_{\alpha,\beta}$. Note that if $a \in K$ then $\phi(a) = a$ since $\psi_{\alpha,\beta}(a) = a$ for $a \in K$. From this it follows that for any of the zeros α_j of $f(x)$ we have:

$$(1.2) \quad f(\phi(\alpha_j)) = a_0 + a_1\phi(\alpha_j) + \dots + a_n\phi(\alpha_j)^n = \phi(a_0 + \dots + a_n\alpha_j^n) = \phi(0) = 0.$$

That is, ϕ maps any zero of $f(x)$ to a zero of $f(x)$. This implies that for all zeros $\alpha_1, \dots, \alpha_n$ of $f(x)$ we have $\phi(\alpha_1), \dots, \phi(\alpha_n) \in E_f$ since E_f contains all zeros of $f(x)$. That is, ϕ maps each α_j into E_f but also maps K into K . Combined this implies that ϕ maps E_f into $E_f = K(\alpha_1, \dots, \alpha_n)$. That is, $\phi : E_f \rightarrow E_f$ is an isomorphism. But now, $\phi(\alpha) = \psi_{\alpha,\beta}(\alpha) = \beta \in E_f$, and since β was an arbitrary conjugate of α this implies that every conjugate of α lies in E_f , and in extension that E_f is a finite normal extension of K . ■

A consequence of the proof of the previous theorem is the following important result.

Lemma 1.1. *Let K be a number field and E/K a finite normal extension of K . If $\alpha \in E$ and β is a conjugate of α then there is some $\psi \in \text{Gal}(E/K)$ such that $\psi(\alpha) = \beta$.*

Proof. Let $f_\alpha(x) \in K[x]$ be the irreducible polynomial for $\alpha \in E$ over K . Let β be a conjugate of α (that is, a zero for $f_\alpha(x)$), since E is a finite normal extension of K we know that $\beta \in E$. We let $\psi_{\alpha,\beta} : K(\alpha) \rightarrow K(\beta)$ be the isomorphism that maps α to β , we can then extend $\psi_{\alpha,\beta}$ to an isomorphism $\psi : E \rightarrow E'$ such that $\psi|_{K(\alpha)} = \psi_{\alpha,\beta}$. We are done if we can show that $E' = E$ since then $\psi(\alpha) = \beta$ and since $\psi_{\alpha,\beta}$ fix K it follows that ψ fix K , so if $E' = E$ then $\psi \in \text{Gal}(E/K)$. If $\gamma \in E$ is any element and if $f_\gamma(x)$ is the irreducible polynomial for γ over K , then $\psi(\gamma)$ is a zero of $f_\gamma(x)$ (try to prove this! It is the same calculation as in the previous lemma). Since $f_\gamma(x)$ splits in E (since E is a finite normal extension) and since $\psi(\gamma)$ is a zero for $f_\gamma(x)$, it follows that $\psi(\gamma) \in E$ and we conclude that $\psi(\gamma) \in E$ for all $\gamma \in E$. That is, $\psi(\gamma) \in E$ for all $\gamma \in E$ which implies $E' = \text{Im}(\psi) = E$. As noted above, this implies that $\psi \in \text{Gal}(E/K)$ and since $\psi(\alpha) = \beta$ we are done. ■

1.2. The fundamental theorem of Galois theory. We end these notes by stating the fundamental theorem of Galois theory, which will be key when proving that one can not solve the general fifth order equation in radicals, and proving a key result which will be used in its proof. In the proof of the fundamental theorem of Galois theory we will use the following theorem, which states that finite extensions of number fields are always simple extensions.

Theorem 1.2 (Primitive element theorem). *Let K be a number field and let E/K be a finite extension of K , then there exists some $\theta \in E$ such that $E = K(\theta)$. In particular, if K is any number field then there is some $\theta \in K$ such that $K = \mathbb{Q}(\theta)$.*

Remark 2. The elements $\theta \in E$ such that $K(\theta) = E$ are called primitive elements, which justifies the name of the theorem.

Proof. Note that if we show that for any two $\alpha, \beta \in \overline{\mathbb{Q}}$ we have some $\theta \in K(\alpha, \beta)$ such that $K(\theta) = K(\alpha, \beta)$ then the general case follows from induction. Indeed, if E is a finite extension of K then $E = K(\alpha_1, \dots, \alpha_n)$ and we can write $K(\alpha_1, \dots, \alpha_n) = (K(\alpha_1, \alpha_2))(\alpha_3, \dots, \alpha_n)$, so if $K(\alpha_1, \alpha_2) = K(\theta_1)$ then we obtain $E = K(\theta_1, \alpha_3, \dots, \alpha_n) = (K(\theta_1, \alpha_3))(\alpha_4, \dots, \alpha_n)$, and then we find θ_2 such that $K(\theta_1, \alpha_3) = K(\theta_2)$. Continuing like this, it follows by induction that there is some θ such that $E = K(\theta)$.

Assume that $\alpha, \beta \in \overline{\mathbb{Q}}$ and consider $E = K(\alpha, \beta)$. For $c \in K$ we let $\theta_c = \alpha + c\beta$. We claim that there is some $c \in K$ such that $E = K(\theta_c)$. Let $\alpha_1, \dots, \alpha_n$ be all the conjugates of α , where $\alpha_1 = \alpha$, and let $\beta_1, \dots, \beta_m$ be all the conjugates of β , where $\beta_1 = \beta$. Notice that the set:

$$(1.3) \quad B = \left\{ \frac{\alpha_i - \alpha_j}{\beta_k - \beta_\ell} \mid k \neq \ell \right\} \subset \overline{\mathbb{Q}}$$

is a finite set since α and β only have finitely many conjugates. Since K is infinite we can choose $c \in K \setminus B$, we will let our c be chosen like this so that $c \notin B$. That is, we let $\theta_c = \alpha + c\beta$ with $c \in K \setminus B$.

Let $f_\alpha(x), g_\beta(x) \in (K(\theta_c))[x]$ be the irreducible polynomials of α and β over $K(\theta_c)$, and let $K(\theta_c) \leq L$ be the splitting field of $f_\alpha(x)g_\beta(x)$. Assume for contradiction that $\beta \notin K(\theta_c)$. Since L is a splitting field of a polynomial with coefficients in $K(\theta_c)$ we know that L is a finite normal extension of $K(\theta_c)$. Note that there has to exist some conjugate $\beta' \in L$ of β such that $\beta' \neq \beta$ since otherwise, if $g_\beta(x)$ only has one zero, then $g_\beta(x) = c(x - \beta)$ with $c \in K(\theta_c)$ which would imply that $\beta \in K(\theta_c)$ but we assumed that this was not the case. So, we can fix a conjugate β' of β such that $\beta' \neq \beta$. By Lemma 1.1 there is $\psi \in \text{Gal}(L/K(\theta_c))$ such that $\psi(\beta) = \beta'$ where $\beta' \neq \beta$. Using that ψ fix $K(\theta_c)$, and in particular satisfies $\psi(\theta_c) = \theta_c$, we have:

$$(1.4) \quad \alpha + c\beta = \theta_c = \psi(\theta_c) = \psi(\alpha) + c\psi(\beta) = \psi(\alpha) + c\beta'$$

and since $\beta \neq \beta'$ we can solve for c and obtain:

$$(1.5) \quad c = \frac{\psi(\alpha) - \alpha}{\beta - \beta'}.$$

Notice that ψ maps α to a conjugate of α over $K(\theta_c)$, since ψ fixes the field $K(\theta_c)$ from which $f_\alpha(x)$ gets its coefficients. It follows that we can write $c \in K$ as a fraction of differences of conjugates of α and β . However, if we can write c as a fraction of differences of conjugates of α and β then $c \in B$, which is a contradiction since we choose $c \notin B$. By this contradiction we conclude that $\beta \in K(\theta_c)$. However, if $\beta \in K(\theta_c)$ then:

$$(1.6) \quad K(\theta_c) \ni \theta_c - c\beta = \alpha + c\beta - c\beta = \alpha$$

so $\alpha \in K(\theta_c)$. Now, finally we have $K(\theta_c) \leq K(\alpha, \beta)$ since $\theta_c = \alpha + c\beta \in K(\alpha, \beta)$ but $\alpha, \beta \in K(\theta_c)$ so we also have $K(\alpha, \beta) \leq K(\theta_c)$. Combined we conclude that $K(\alpha, \beta) = K(\theta_c)$ which finishes the proof of the lemma. $\blacksquare$

We end by stating the fundamental theorem of Galois theory, which we will prove next class. Before stating the theorem we recall some constructions of subgroups of the Galois group, and subfields of a finite normal extension. Let K be a number field and E a finite normal extension of K , then for any subgroup $H \leq \text{Gal}(E/K)$ we define:

$$(1.7) \quad E^H = \{\alpha \in E \mid h(\alpha) = \alpha, \text{ for all } h \in H\}.$$

If $K \leq L \leq E$ is a subfield of E that contains K then:

$$(1.8) \quad H_L = \{h \in \text{Gal}(E/K) \mid h(\ell) = \ell, \text{ for all } \ell \in L\}.$$

The main content of the fundamental theorem of Galois theory is that these constructions are inverse operations. That is, there is a one-to-one correspondence between subgroups of $\text{Gal}(E/K)$ and subfields $K \leq L \leq E$ that contains K .

Theorem 1.3. *Let K be a number field and let E be a finite normal extension of K . The following holds:*

- (i) *For any subfield $K \leq L \leq E$ there is a subgroup $H \leq \text{Gal}(E/K)$ such that $L = E^H$.*
- (ii) *For any subgroup $H \leq \text{Gal}(E/K)$ there is a subfield $K \leq L \leq E$ such that $H_L = H$.*
- (iii) *The correspondence between subgroups and subfields are inverses: $H_{E^H} = H$ and $E^{H_L} = L$.*
- (iv) *A intermediate field extension $K \leq L \leq E$ is a normal extension of K if and only if $H_L \leq \text{Gal}(E/K)$ is a normal subgroup.*
- (v) *If $K \leq L \leq E$ is a finite normal extension of K then*

$$\text{Gal}(L/K) \cong \text{Gal}(E/K)/H_L$$

where the quotient $\text{Gal}(E/K)/H_L$ is a group since H_L is a normal subgroup of $\text{Gal}(E/K)$ by property (iv).

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